

Step 1:

Field	Value
Step	1
Module	RECON
Hypothesis	Identify exposed services and application stack
Action	nmap -sV -sC -p 8081 100.113.75.28
Outcome	Success
Pivot?	No
Notes	Apache 2.4.65 (Debian), PHP, DVWA detected. robots.txt exposes /admin/, /backup/, /debug/

Step 2:

Field	Value
Step	2
Module	AUTH
Hypothesis	Default DVWA credentials may grant admin access
Action	POST login.php with admin/password and CSRF token
Outcome	Success
Pivot?	Yes
Notes	Authentication successful, redirected to index.php

Step 3:

Field	Value
Step	3
Module	ENUMERATION
Hypothesis	Identify available DVWA vulnerability modules
Action	Browse index.php and enumerate /vulnerabilities/ endpoints
Outcome	Success
Pivot?	No
Notes	Modules discovered: SQLi, File Upload, File Inclusion, Exec, XSS, Open Redirect

Step 4:

Field	Value
Step	4
Module	SQLi
Hypothesis	SQL injection may reveal database contents or flag
Action	Manual SQLi attempts + debug=1 + skipwaf testing
Outcome	Partial Success
Pivot?	Yes
Notes	Hidden parameter skipwaf=1 identified but exploitation deprioritized

Step 5:

Field	Value
Step	5
Module	UPLOAD
Hypothesis	File upload module may allow PHP webshell upload
Action	Craft PHP webshell with GIF89a magic bytes; upload via POST
Outcome	Success
Pivot?	Yes
Notes	File stored under /hackable/uploads/.../img_XXXX.gif

Step 6:

Field	Value
Step	6
Module	RCE
Hypothesis	Uploaded polyglot GIF/PHP file will execute when accessed via URL
Action	Access uploaded file with ?cmd=id
Outcome	Success
Pivot?	No
Notes	uid=33(www-data) confirmed — Remote Code Execution achieved

Step 7:

Field	Value
Step	7
Module	POST-EXPLOIT
Hypothesis	Flag stored in /foothold/flag.txt
Action	Use PHP runtime via webshell to read /foothold/flag.txt
Outcome	Success
Pivot?	No
Notes	FLAG{INITIAL_FOOTHOLD_ACHIEVED} retrieved

Metrics:

Total steps: 7

Total pivots: 2

Dead-end modules: 0

Exploit chain depth: 5